

Sec 4.2 – Optimization

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

The largest (or smallest) value of a function f over a specified domain is called the **global maximum** (or **minimum**) of f . In Sec 4.1 we considered the **local maximum** and **minimum**, which as the name implies gives the largest or smallest value is locally.

Definition

Suppose p is a point in the domain of f :

- f has a **global minimum** at p if $f(p)$ is less than or equal to all values of f .
- f has a **global maximum** at p if $f(p)$ is greater than or equal to all values of f .

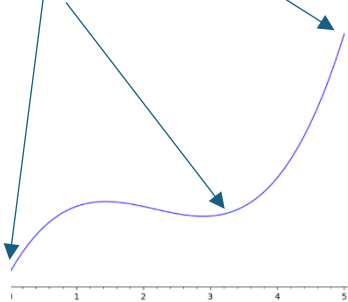
Optimization is the process of finding the global maximum or minimum, the *optimal* result for a problem.

Notes:

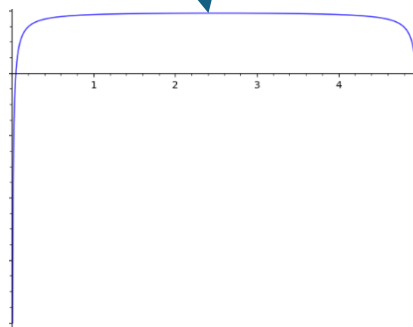
- Global maxima and minima are also referred to as *absolute* maxima and minima.
- The global maximum and minimum are values of the function and refer to the *y-values* of the function, while local minimums and local maximums usually refer to the points, the *x-values*, since there can be several *x-values*.

For both sets of characteristics below, draw a function on $[0,5]$:

With both a Global Maximum and Global Minimum, and one additional Local Minimum.



With a Global Maximum, but no Global Minimum ← **Vertical Asymptotes**



Theorem (Extreme Value Theorem)

If f is continuous on the closed interval $a \leq x \leq b$, then f has a global maximum and a global minimum on that interval.

Finding global maxima and minima on a closed interval

For a continuous function f on a closed interval $a \leq x \leq b$:

- Find the critical points of f in the interval.
- Evaluate the function at the critical points and at the endpoints, a and b .

The largest (y -)value of the function is the global maximum and the smallest (y -)value is the global minimum.

So, this process is different than what we did in **Sec 4.1 to find Local Extrema:**

- Find critical points
- Test if they are Local-Max or Local-Min
 - 1st Derivative Test
 - 2nd Derivative Test

Here, in **Sec 4.2 to find Global Extrema:**

- Find critical points
- Evaluate the **original function** at all critical points and endpoints
- The largest and smallest values are the Global-Max and Global-Min

Find the global maximum and minimum for $g(x) = -\frac{1}{3}x^3 - 2x^2 + 5x$ on the interval $[-3, 2]$.

Solution:

$g'(x) = -x^2 - 4x + 5 = -(x^2 + 4x - 5) = -(x + 5)(x - 1)$. We set this equal to zero to find critical points and the zeros are $x = -5, 1$. The only critical point in the interval is $x = 1$.

Now we check $g(x)$ at the endpoints and at the critical points:

$$g(-3) = -24$$

$$g(1) = 2\frac{2}{3}$$

$$g(2) = -\frac{2}{3}$$

Therefore, the global minimum is -24 at $x = -3$ and the global maximum is $2\frac{2}{3}$ at $x = 1$.

So what happens if we have a function defined on an **open** interval?

Finding global maxima and minima on an open interval.

For a continuous function f on an open interval:

- Find the critical points of f in the interval.
- Evaluate the function at all the critical points.
- Use information from f' and f'' or limits to determine how f changes outside the critical points (when x approaches the endpoints of the interval).
- The largest (y -)value of the function is the global maximum and the smallest (y -)value is the global minimum.

Find the global maximum and minimum for $f(x) = 3x^4 + 4x^3 - 12x^2 - 10$ on the real numbers.

Solution:

$$f'(x) = 12x^3 + 12x^2 - 24x = 12x(x^2 + x - 2) = 12x(x + 2)(x - 1).$$

The critical points are $x = -2, 0, 1$.

We check values at these points and then consider what is happening as $x \rightarrow \pm\infty$. $f(-2) = -42$

$$f(0) = -10$$

$$f(1) = -15$$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ (you can see this either by finding $\lim_{x \rightarrow -\infty} f(x)$ or by noting that $f'(x) < 0$ for any $x < -2$.)

Similarly, as $x \rightarrow \infty$, $f(x) \rightarrow \infty$.

Therefore, there is no global maximum since the values go to ∞ . There is a global minimum at -42 when $x = -2$.

Partner up

A **parameter** is a constant that depends on the situation and frequently the maximum or minimum depends on the parameter.

Problem 4.2 #30: For some positive constant C , a patient's temperature change, T , due to a dose, D , of a drug is given by

$$T = \left(\frac{C}{2} - \frac{D}{3} \right) D^2.$$

(a) What dosage maximizes the temperature change (for a given C).

(a) We view T as a function of D , so we take the derivative with respect to D , treating C as the constant it is:

First simplify the function: $T(D) = \left(\frac{C}{2} - \frac{D}{3}\right) D^2 = \frac{C}{2} D^2 - \frac{1}{3} D^3$.

Then the derivative is $\frac{dT}{dD} = T'(D) = CD - D^2 = (C - D)D$

The critical values are when $(C - D)D = 0$, or when $D = 0, C$.

The values at the critical points are $T(0) = 0$

$T(C) = \left(\frac{C}{2} - \frac{C}{3}\right) C^2 = \frac{C^3}{6}$.

This function is on the open interval $(0, \infty)$, so we consider what occurs as $D \rightarrow \infty$: the term $C - D$ is negative and D is positive, so $T'(D) < 0$.

This means that $T(C) = \frac{C^3}{6}$ is the maximum.